Guaranteeing Physical Validity by Construction: A Systematic Factorial Evaluation of Hard vs. Soft Thermodynamic Constraint Enforcement in Physics-Informed Neural Networks for Sorption Studies

Abstract

To be written after the main results and discussion have been finalized.

1 Introduction

To be written.

2 Materials and Methods

2.1 Dataset and prediction task

The study considers methane–coal sorption data under different pressure, temperature, and coal-composition conditions. The raw dataset contains 993 adsorption measurements. Because the CSV did not include experiment identifiers, experiment groups were constructed from rows with the same coal composition and temperature, resulting in 114 experiment groups. Rows in the same experiment group are treated as one adsorption curve, meaning a set of pressure–adsorption measurements collected at fixed coal composition and temperature. Splitting was performed at the experiment level to avoid placing points from the same curve in both training and test sets.

The raw input columns are moisture content, ash content, volatile matter, temperature, and pressure. The prediction target is adsorption capacity. The corresponding variables used by the model are summarized as follows:

Role	**Model variable**
Temperature input	`temperature_K`
Pressure input	`pressure_MPa`
Moisture input	`moisture_wt`
Ash input	`ash_wt`
Volatile-matter input	`volatile_wt`
Prediction target	`CH4_adsorption_m3t`

Since thermodynamic formulas require absolute temperature, the measured values were converted from degrees Celsius to Kelvin before any thermodynamic feature or constraint was evaluated. Since the Sips formulation requires raw positive pressure, all samples were checked to satisfy $P > 0$. Although adsorption capacity is physically non-negative, the dataset was checked to ensure strictly positive target values, since the logarithmic target transform requires $q > 0$.

The raw CSV did not include a measured `coal_rank` column. To define the T2 coal-rank extrapolation tier, coal-rank labels were constructed from the available coal-composition measurements: moisture, ash, and volatile matter. Volatile matter and fixed carbon were first

expressed on a dry, ash-free basis, and threshold rules were then used to assign each sample to one coal rank. This produced four ranks in the current dataset: anthracite, low-volatile bituminous, medium-volatile bituminous, and subbituminous/lignite. The constructed coal-rank label is used only for T2 grouping and interpretation analyses, not as a model input.

2.2 Preprocessing and feature engineering

Preprocessing makes the data valid, comparable, and suitable for training. Engineered features were also included to provide useful derived information to the model. The model input vector contains 12 numerical features:

$$\mathbf{x} = [\text{moisture}, \text{ash}, \text{volatile}, T, P, T_r, P_r, \text{FC}, \text{OM}, \beta, PT, \text{MV}].$$

Here,

$$\begin{aligned} T_r &= T/190.6, \\ P_r &= P/4.60, \\ \text{FC} &= 100 - \text{moisture} - \text{ash} - \text{volatile}, \\ \text{OM} &= (\text{volatile} + \text{FC})/100, \\ \beta &= (RT)^{-1}, \\ PT &= PT, \\ \text{MV} &= \text{moisture} \times \text{volatile}. \end{aligned}$$

The gas constant was set to $R = 8.314\ \text{J mol}^{-1}\,\text{K}^{-1}$.

All input features were scaled using only the training data, so that the model could train on variables with comparable numerical ranges without using information from the validation or test sets. The adsorption target was log-transformed during training, and predictions were converted back to the original adsorption scale before computing the reported error metrics. Raw pressure in MPa was also kept separately, because the Sips-based model components and physical checks require the real pressure values rather than the scaled pressure feature.

2.3 Leakage-safe data splitting

To avoid data leakage, all splits were performed at the experiment-group level rather than at the individual-row level. Rows belonging to the same adsorption curve share the same `experiment_id`, and were therefore kept together in the same split. This prevents the model from being trained on part of an adsorption curve and then tested on another pressure point from the same curve.

The data pipeline defines three evaluation tiers. T1 is the interpolation test set, containing held-out experiment groups from coal ranks that are also present in training. T2 is the coal-rank extrapolation set, where one coal-rank group is withheld from training. T3 is the pressure extrapolation set, defined by high-pressure samples inside the T1 test set. Cross-validation was also performed with GroupKFold using `experiment_id` as the grouping variable, so that validation folds remain leakage-safe.

2.4 Soft-constrained baseline

The soft-constrained baseline was implemented as the reference model for the study. It uses the 12 engineered input features to predict methane adsorption capacity, while also estimating auxiliary physical quantities used in the physics penalties. Physical consistency is encouraged through additional loss terms for saturation bounds, pressure monotonicity, Sips consistency, and temperature dependence, but these constraints are not guaranteed by the model architecture.

A sweep over the physics-loss weight λ_p was performed to examine the trade-off between predictive accuracy and physical consistency. The trained baseline was evaluated on the T1 interpolation, T2 coal-rank extrapolation, and T3 pressure-extrapolation tiers, and serves as the reference for comparison with the hard-constrained variants.

2.5 Hard-constrained model variants

Hard-constrained model variants were implemented to compare against the soft-constrained baseline and to test whether direct architectural enforcement improves physical validity and extrapolation performance. In this project, the hard variants are:

V3	hard bounds,
V4	hard pressure monotonicity,
V5	hard Sips gauge,
V6	hard Van't Hoff/Arrhenius structure,
V7	joint hard bounds and hard monotonicity,
V2/V8	all-hard combined variants.

The hard-bounds variant enforces the adsorption range directly,

$$0 < \hat{q} < \hat{q}_{\max}.$$

Instead of penalizing saturation violations after they occur, the output head is designed so that the adsorption prediction remains between zero and the learned maximum capacity.

The hard-monotonicity variant enforces non-decreasing adsorption with respect to pressure,

$$\frac{\partial \hat{q}}{\partial P} \geq 0.$$

Pressure enters through a monotone pathway with non-negative weights and monotone activations. This makes the pressure contribution unable to decrease the predicted adsorption. In the implementation, raw pressure and the pressure-derived engineered features P_r and PT were excluded from the unconstrained composition branch. This is necessary because an unconstrained branch could otherwise learn a decreasing pressure effect and break the monotonicity guarantee.

The hard Sips-gauge variant does not predict adsorption as a completely free output. Instead, it predicts the Sips parameters

$$\hat{q}_{\max}, \quad \hat{K}, \quad \hat{n},$$

from coal-composition features, and then computes adsorption using raw pressure:

$$\hat{q}(P) = \hat{q}_{\max} \frac{(\hat{K}P)^{\hat{n}}}{1 + (\hat{K}P)^{\hat{n}}}.$$

This gives the model a structured pressure–adsorption relation.

The hard Van't Hoff/Arrhenius variant constrains the temperature dependence of the adsorption constant. The model predicts an Arrhenius-type structure so that the relationship between K, temperature, and adsorption enthalpy ΔH is built into the model. Finally, the joint hard variants combine multiple hard constraints to test whether the physical rules work well together or whether they reduce accuracy or extrapolation performance.

2.6 Training protocol

All model variants were trained using the same general training protocol. The training data were provided in mini-batches with a default batch size of 64. For each batch, the model predicted

methane adsorption capacity and the relevant auxiliary physical quantities. The total loss was then computed from the data-fitting term and, where applicable, the physics-consistency terms.

Model parameters were optimized using the AdamW optimizer with learning rate 10^{-3} and weight decay 10^{-5}. The full V1 physics-weight sweep used the intended network architecture with hidden dimensions $(512, 1024, 512, 256, 128)$, a maximum of 1000 epochs, early stopping patience of 100 epochs, three random seeds, and five grouped cross-validation folds. The physics-loss weights were $\lambda_p \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0, 2.0\}$, giving 15 validation runs per value of λ_p.

Gradient clipping with maximum norm 5 was used to improve numerical stability.

During training, the validation set was used only for model selection and early stopping. The model state with the lowest validation loss was restored after training and saved as a checkpoint. Final evaluation was then performed on the held-out T1, T2, and T3 evaluation tiers using the unified evaluation harness.

2.7 Evaluation metrics and statistical tests

All trained models were evaluated using the same unified evaluation harness and the same T1, T2, and T3 evaluation tiers. Predictive accuracy was measured on the original adsorption scale after reversing the logarithmic target transform. The main accuracy metrics were the coefficient of determination R^2, root mean squared error (RMSE), mean absolute error (MAE), maximum absolute error (MaxAE), and relative error.

Physical consistency was evaluated using violation rates. These included the fraction of negative adsorption predictions, the fraction of predictions above the predicted saturation capacity $q_{\max}$, and the fraction of pressure–adsorption monotonicity violations. These metrics were used to assess whether a model only fit the data numerically or also respected the intended physical behavior.

Uncertainty quality was evaluated using Monte Carlo dropout predictions and calibration metrics, including empirical interval coverage and expected calibration error (ECE). Confidence intervals for accuracy and physical violation metrics were estimated by bootstrap resampling at the experiment level, preserving the grouped structure of the data.

For model comparisons, paired statistical tests were performed at the experiment level. In particular, pairwise RMSE differences between model variants were computed on the same evaluation samples, and one-sample tests were applied to the experiment-wise differences. When multiple pairwise comparisons were made, Bonferroni correction was applied to control for multiple testing.

2.8 Uncertainty calibration

Uncertainty calibration was used to assess whether the predictive uncertainty estimates were reliable. For each trained model, Monte Carlo dropout was used to generate repeated stochastic predictions for the same input samples. The spread of these predictions was used as an estimate of epistemic uncertainty, while the model-predicted variance represented aleatoric uncertainty when available.

Calibration was evaluated by comparing nominal prediction intervals with their empirical coverage. For example, a well-calibrated 90% prediction interval should contain the observed adsorption value for approximately 90% of the samples. Expected calibration error (ECE) was computed as the average difference between nominal and empirical coverage across several interval levels.

A scalar temperature-scaling factor was selected on the validation set to adjust the predicted uncertainty widths. The calibrated uncertainty estimates were then evaluated on the T1, T2, and T3 tiers. This allowed us to test whether the models became more uncertain

under extrapolation, especially for the coal-rank-extrapolation and high-pressure-extrapolation settings.

2.9 Interpretability and physical validation

Interpretability and physical validation were used to examine whether the trained models learned physically meaningful relationships. This included feature importance patterns, Sips parameter trends, and adsorption enthalpy behavior across coal ranks. Feature importance was evaluated using SHAP-style analyses for the baseline and Sips-gauge models. The learned Sips parameters $\hat{q}_{\max}$, $\hat{K}$, and $\hat{n}$ were summarized by coal rank and compared with classical Langmuir fits. The learned adsorption enthalpy ΔH from the Arrhenius variant was also analyzed by coal rank to check whether the inferred temperature dependence was physically plausible.

3 Results and Discussion

3.1 Evaluation setup and data splits

The processed dataset contained 993 adsorption measurements grouped into 114 constructed experiment groups. The leakage-safe splitting procedure assigned complete experiment groups to each split, so no constructed experiment identifier was shared between training, validation, and held-out evaluation data. After withholding the T2 coal-rank extrapolation set, the remaining data were split into training, validation, and T1 interpolation test sets. The final split sizes are shown in Table 1.

Split	**Samples**	**Experiments**
Full dataset	993	114
Training pool excluding T2	752	86
Training	472	54
Validation	132	14
T1 interpolation	148	18
T2 coal-rank extrapolation	241	28
T3 pressure extrapolation	3	2

Table 1: Leakage-safe data splits used for training, validation, and held-out evaluation.

The T2 tier consisted of all samples inferred as subbituminous/lignite, giving a held-out coal-rank extrapolation test with 241 samples from 28 experiments. The T3 pressure-extrapolation tier contained samples from the T1 set with pressure above 8 MPa. Because only three samples satisfied this condition, T3 should be interpreted with caution. Across the full dataset, the pressure range was 0.0019342–9.2587632 MPa, and all pressure values were strictly positive.

3.2 Soft baseline and physics-loss weighting

The soft-constrained baseline was designed to predict methane adsorption while encouraging physical consistency through additional penalty terms. The total training objective combines a data-fitting loss with a physics loss,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{physics}},$$

where λ_p controls the strength of the physics regularization. A larger value of λ_p gives more importance to the physical constraints, whereas $\lambda_p = 0$ corresponds to a purely data-driven baseline.

The implemented physics loss includes terms related to adsorption bounds, pressure monotonicity, Sips consistency, and temperature dependence. The purpose of the λ_p sweep is therefore to identify a balance between numerical accuracy and physical consistency. In practical terms, this analysis asks whether the model can fit the measured adsorption values while also reducing physically implausible behavior.

The full V1 sweep is shown in Table 2. Model selection used a validation score that combines accuracy and physical validity,

$$s_{\mathrm{val}} = \mathrm{RMSE}_{\mathrm{val}} \left(1 + v_{\mathrm{val}}\right),$$

where v_{val} is the largest validation violation rate among negative adsorption, saturation-bound, and non-monotonicity checks. The unregularized model ($\lambda_p = 0$) had a competitive RMSE but a violation rate of 1.0. The selected value was $\lambda_p = 0.05$, which gave the lowest combined validation score while keeping the violation rate close to zero.

λ_p	Val. score	Val. RMSE	Val. violation	Val. ECE	Mean epochs
0.00	10.349	5.174	1.000	0.034	116.1
0.01	5.041	5.018	0.004	0.049	154.5
0.05	**4.916**	**4.897**	0.004	0.050	181.6
0.10	5.050	5.002	0.010	0.052	213.4
0.20	5.036	5.004	0.006	0.055	243.9
0.50	4.969	4.966	**0.001**	0.056	315.3
1.00	4.941	4.922	0.004	0.049	428.6
2.00	5.843	5.840	0.001	0.064	445.5

Table 2: Full V1 physics-weight sweep using three seeds and five grouped cross-validation folds. The selected value is $\lambda_p = 0.05$.

3.3 Comparison of soft and hard variants

The soft baseline and hard-constrained variants were compared across the three evaluation tiers using the same evaluation harness. Table 3 summarizes the full-protocol RMSE values. Lower RMSE indicates smaller prediction error on the original methane adsorption scale.

Variant	T1 RMSE	T2 RMSE	T3 RMSE
V1 soft baseline	5.300	5.746	4.943
V3 hard bounds	5.624	5.903	2.217
V4 hard monotonicity	**5.213**	5.709	4.643
V5 hard Sips gauge	5.410	5.476	1.372
V6 hard Van't Hoff / Arrhenius	6.082	5.543	5.478
V7 hard bounds + monotonicity	7.116	6.145	**1.192**
V2/V8 all-hard combined	6.209	**5.056**	1.435

Table 3: Full-protocol RMSE values for the soft baseline and hard-constrained variants on the T1, T2, and T3 evaluation tiers.

For the full saved runs, the hard monotonicity variant gives the lowest T1 RMSE, although its advantage over the soft baseline is small. On the T2 coal-rank extrapolation tier, the V2/V8 all-hard combined variants give the lowest RMSE. This suggests that combining the Sips-gauge and Van't Hoff / Arrhenius structures can help when the model is evaluated on a coal rank withheld from training. On the T3 pressure-extrapolation tier, the joint hard-bounds and

monotonicity variant gives the lowest RMSE, but this result should be interpreted carefully because T3 contains only three samples.

These results illustrate the main comparison that the hard-variant analysis is designed to answer: different physical constraints may help in different evaluation regimes. A constraint that improves interpolation does not necessarily give the best extrapolation result, and combining constraints can improve one tier while reducing performance on another. The strongest pattern is therefore conditional: hard constraints help most clearly for physical validity and for some extrapolation tiers, but they do not dominate every accuracy metric.

3.4 Physical validity of constrained architectures

Physical validity was evaluated separately from predictive accuracy. This is important because a model can have a low numerical error while still producing physically implausible behavior. The main physical checks were non-negative adsorption, adsorption below the predicted saturation capacity, non-decreasing adsorption with pressure, preservation of the Sips structure, and consistency with the Arrhenius/Van't Hoff temperature relation.

Variant	**Physical check**	**Current result**
V4 hard monotonicity	$\partial\hat{q}/\partial P \geq 0$	minimum slope = 0.111, violation rate = 0
V5 hard Sips gauge	Sips bounds and monotonicity	lower, upper, and slope violation rates = 0
V6 hard Van't Hoff / Arrhenius	Arrhenius identity	maximum identity error $= 1.82 \times 10^{-12}$
V7 hard bounds + monotonicity	$\partial\hat{q}/\partial P \geq 0$	minimum slope = 0.146, violation rate = 0
V2/V8 all-hard combined	Sips bounds and monotonicity	lower, upper, and slope violation rates = 0

Table 4: Architecture-level physical verification checks for the currently saved hard-constrained variants.

The hard monotonicity variants passed the pressure-monotonicity verification: the computed pressure derivatives were non-negative on the tested pressure grid. The hard Sips-gauge variants also satisfied the intended Sips structure. For V5, the minimum predicted adsorption was positive ($\min \hat{q} = 0.0035$), the minimum margin $\hat{q}_{\max} - \hat{q}$ was positive (0.029), and no lower-bound, upper-bound, or negative-slope violations were detected. This confirms that the Sips-gauge architecture enforces the desired pressure–adsorption shape by construction, rather than relying only on a penalty term during training.

The Arrhenius variants satisfied the algebraic Arrhenius identity to numerical precision. However, the learned adsorption enthalpy values require more careful interpretation. For V6, the mean learned adsorption enthalpy was approximately -44.2 kJ mol^{-1}, and about 76% of sampled values lay in the expected range used by the verification script. For the V2/V8 all-hard combined variant, the Arrhenius identity was also satisfied, but the sampled enthalpy values did not fall in that expected range. This distinction is important: satisfying the mathematical form of a constraint does not automatically guarantee that all learned physical parameters are scientifically plausible.

Overall, the physical-validity checks show the difference between soft and hard constraints. Soft constraints discourage violations through the loss function, whereas hard constraints remove certain violations from the model class itself. The full-run verification confirms that the implemented hard architectures enforce their intended structural properties.

3.5 Ablation and interaction effects

The ablation analysis was used to separate the effect of individual hard constraints from the effect of combining them. Two factorial comparisons were implemented. The first comparison studies hard bounds and hard monotonicity using V1, V3, V4, and V7. The second comparison studies the Sips-gauge and Van't Hoff / Arrhenius structures using V1, V5, V6, and V8. In each case, the analysis compares the unconstrained or soft reference, the two single-constraint variants, and the joint hard-constraint variant.

Family	Tier	A only	B only	Joint	Interaction
Bounds + monotonicity	T1	5.624	5.213	7.116	antagonistic
Bounds + monotonicity	T2	5.903	5.709	6.145	antagonistic
Bounds + monotonicity	T3	2.217	4.643	1.192	superadditive
Sips + Van't Hoff	T1	5.410	6.082	6.209	antagonistic
Sips + Van't Hoff	T2	5.476	5.543	5.056	superadditive
Sips + Van't Hoff	T3	1.372	5.478	1.435	superadditive

Table 5: Full-run Step 9 interaction summary using RMSE values. The interaction label is computed from the factorial contrast in the ablation script.

The full-run ablation results indicate that single constraints and joint constraints do not behave identically across the evaluation tiers. For T1, the single hard monotonicity model has the lowest RMSE, whereas the V2/V8 all-hard combined variants give the lowest T2 RMSE. This suggests that the most useful constraint can depend on whether the model is being tested in T1 interpolation or under coal-rank extrapolation. The interaction results are also tier-dependent: bounds and monotonicity are antagonistic on T1 and T2 but superadditive on T3. The Sips and Van't Hoff combination is antagonistic on T1 and superadditive on T2 and T3. The T3 findings remain uncertain because T3 contains only three samples.

The pairwise statistical analysis gives the same warning. In the regenerated Step 9 results, 2 of 28 pairwise RMSE comparisons were significant after Bonferroni correction on T1, and 7 of 28 were significant on T2. No pairwise RMSE comparisons were significant on T3 after correction. This supports using T3 mainly as qualitative pressure-extrapolation evidence rather than as strong statistical evidence.

3.6 Uncertainty calibration

Uncertainty calibration was evaluated using 100 Monte Carlo dropout samples per model and split. For each model, a scalar calibration factor τ was fitted on the validation set. This factor rescales the predicted standard deviation: $\tau = 1$ leaves the uncertainty unchanged, while $\tau > 1$ makes the prediction intervals wider. Calibration quality was measured using expected calibration error (ECE), where smaller values indicate that the empirical coverage of prediction intervals is closer to the nominal confidence levels.

The full saved results show that validation calibration adjusts the uncertainty estimates for most variants. The V7 hard-bounds and monotonicity model obtains the lowest validation ECE, while the V2/V8 all-hard combined variants obtain the lowest T1 and T2 ECE values in the current table.

The main pattern is that calibration on the validation set does not always transfer to extrapolation. Several hard variants have low T1 ECE but much higher T2 ECE. This means that their uncertainty intervals are reasonably calibrated for T1 interpolation samples, but less reliable for the withheld coal-rank extrapolation tier. In the full results, the V2/V8 all-hard combined variants give the smallest T2 ECE, suggesting that combining the hard Sips and hard

Variant	τ	Val. ECE	T1 ECE	T2 ECE	T3 ECE
V1 all-soft	1.059	0.032	0.027	0.273	0.120
V2 all-hard	1.122	0.026	**0.015**	**0.202**	0.120
V3 hard bounds	1.334	0.042	0.053	0.242	0.120
V4 hard monotonicity	0.891	0.039	0.026	0.321	0.120
V5 hard Sips	1.122	0.041	0.032	0.276	0.120
V6 hard Van't Hoff / Arrhenius	1.259	0.045	0.036	0.308	0.120
V7 hard bounds + monotonicity	1.000	**0.017**	0.015	0.299	0.120
V8 hard Sips + Van't Hoff / Arrhenius	1.122	0.026	**0.015**	**0.202**	0.120

Table 6: Full-run calibrated ECE values for validation, T1 interpolation, T2 coal-rank extrapolation, and T3 pressure extrapolation. Lower ECE indicates better uncertainty calibration.

Van't Hoff / Arrhenius structures gives the most stable uncertainty behavior under coal-rank extrapolation.

The epistemic uncertainty fraction also varies between variants. In general, larger uncertainty under the held-out tiers is consistent with the idea that the model expresses more model-form uncertainty when predictions are made farther from the training distribution. However, because T3 contains only three samples, the T3 uncertainty results should be treated as qualitative evidence only.

3.7 Interpretability and physical validation

Interpretability analysis was used to check whether the trained models relied on physically meaningful variables and whether the learned physical parameters followed reasonable coal-rank trends. The analysis included SHAP-style feature importance, learned Sips-parameter summaries, comparison with classical Langmuir fits, and adsorption enthalpy summaries from the Van't Hoff / Arrhenius model.

Rank	V1 top feature	V5 top feature
1	Pr	pressure_MPa
2	FC	FC
3	PT	volatile_wt
4	MV	MV
5	pressure_MPa	Tr

Table 7: Top five global feature-importance rankings for the soft baseline and hard Sips-gauge model in the current Step 11 outputs.

For the hard Sips-gauge model, pressure was the most important input feature, followed by fixed carbon and volatile matter. This is physically sensible because the Sips-gauge architecture computes adsorption as a function of raw pressure and learned coal-dependent Sips parameters. The V1 and V5 feature rankings were weakly and not significantly aligned in the current outputs ($\rho_s = 0.231$, $p = 0.471$), showing that the hard Sips structure changes how the model distributes importance across pressure, temperature, and composition features.

The learned Sips parameters show a clear coal-rank pattern. The mean $\hat{q}_{\max}$ is highest for anthracite and lower for lower-rank coal. Across all 114 experiments, $\hat{q}_{\max}$ was negatively correlated with volatile matter ($r = -0.687$, $p = 3.04\times10^{-17}$), satisfying the blueprint check that higher volatile matter should generally correspond to lower inferred adsorption capacity. The comparison with classical Langmuir fits also supported this interpretation: all 114 experiment-level Langmuir fits succeeded, and V5 $\hat{q}_{\max}$ correlated with the fitted Langmuir capacity q_L ($r = 0.739$, $p = 6.11 \times 10^{-21}$).

Split	Coal rank	$\hat{q}_{\max}$	$\hat{n}$	ΔH
Train	Anthracite	24.66	1.53	-45.52
Train	Low-volatile bituminous	20.80	1.22	-46.00
Train	Medium-volatile bituminous	15.70	0.98	-35.54
T2	Subbituminous/lignite	10.25	0.75	-25.09

Table 8: Selected coal-rank summaries of learned physical quantities. $\hat{q}_{\max}$ and $\hat{n}$ are from V5, while ΔH is from V6 and is reported in $\mathrm{kJ\,mol^{-1}}$.

The Van't Hoff / Arrhenius model produced negative adsorption enthalpy values, consistent with exothermic adsorption. Mean ΔH values were inside the expected reference range used by the verification script for all reported ranks. The rank effect was statistically significant for the training split, validation split, and T1 split. For T2 and T3, rank-wise statistical comparison was not possible because each contained only one inferred rank in the current split definition.

Overall, the interpretability results support the physical reading of the hard models: pressure controls the adsorption curve, coal composition affects the learned capacity, and the inferred thermodynamic quantities vary with coal rank. The current Step 11 outputs show that the hard-constraint models are inspectable in physically meaningful terms.

3.8 Summary of findings

The current analysis establishes the full evaluation structure for the study. The dataset was split at the experiment level, with separate tiers for T1 interpolation, T2 coal-rank extrapolation, and T3 pressure extrapolation. This makes it possible to evaluate not only whether a model fits known adsorption behavior, but also whether it behaves reasonably when the test data move away from the training distribution.

The hard-constrained models show the strongest numerical performance on several evaluation tiers, but not uniformly across all tiers. The hard monotonicity model gives the lowest T1 RMSE, the V2/V8 all-hard combined variants give the lowest T2 RMSE, and the joint hard-bounds and monotonicity model gives the lowest T3 RMSE. The T3 result is only qualitative because the T3 set contains three samples. The physical-verification checks show that the hard architectures enforce their intended structural properties, including non-negative adsorption, saturation bounds, and pressure monotonicity.

The ablation analysis suggests that physical constraints do not contribute in a single uniform way. Some constraints help more for interpolation, while others are more useful for extrapolation. The uncertainty analysis shows that calibration fitted on the validation set does not always transfer cleanly to the coal-rank extrapolation tier. The interpretability analysis supports the physical reading of the models: pressure and fixed carbon are important features, learned Sips capacities vary with coal rank, and the Van't Hoff / Arrhenius model produces negative adsorption enthalpy values consistent with exothermic adsorption.

The main finding is that hard constraints improve physical validity by construction, while their effect on predictive accuracy and calibration depends on the evaluation tier. The full V1 tuning protocol strengthens the comparison with the soft baseline, but the small T3 sample size remains an important limitation.

4 Conclusions

The full saved results show that hard thermodynamic constraints provide a clear advantage in physical validity, since the constrained architectures remove specific classes of violations by construction. However, this advantage does not translate into uniformly better predictive accuracy or uncertainty calibration. The best-performing model depends on the evaluation

tier: the hard monotonicity model performs best on T1, the V2/V8 all-hard combined variants perform best on T2, and the joint bounds–monotonicity model performs best on T3, although T3 is too small for strong statistical claims. Thus, hard constraints outperform soft penalties most clearly in guaranteed physical validity, while their accuracy and calibration benefits are conditional and come at the cost of additional architectural complexity and computational expense.